

1.  CARDOSO-SILVA et al. (2025). The GENIAL Framework.

Mapping Student-GenAI Interactions onto Experiential Learning: The GENIAL Framework

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ABSTRACT

This study proposes a conceptual framework for understanding how generative artificial intelligence (GenAI) affects university students' learning journey. The use of GenAI tools in academic settings left scholars at a lack of theoretical tools to capture the impact of how chatbots impact learning. Current studies rely heavily on self-reports and controlled experiments, missing how students engage with chatbots while learning. Building on literature on experiential learning and styles of student engagement with GenAI as theoretical underpinnings, the GENIAL framework sheds light on the dynamics of AI-human interaction in all stages of the experiential learning cycle, allowing us to identify how the cycle of learning is affected by the use of GenAI tools. Our findings, based on the analysis of 200 course chatlogs at a London-based research-intensive university and two case studies in which we illustrate the use of the GENIAL framework, reveal that GenAI can catalyse or disrupt learning depending on student agency and contextual factors. Our framework reveals how students learn with GenAI and identifies if critical steps are missed. By introducing a diagnostic notation that maps GenAI interactions onto Kolb's learning stages, the framework enables educators to mark whether each stage is catalysed (+), disrupted (-), or skipped entirely. We contribute to the literature by turning vague concerns about 'AI cheating' into precise diagnostic evaluation that enables educators and policymakers to address the negative impacts of GAI usage on learning as well as the curriculum.